

Bachelors with Mathematics as Major 5th Semester MMT522J3: Mathematics/Applied Mathematics: FOURIER AND LAPLACE TRANSFORM	
Credits: 4 THEORY + 2 TUTORIAL	Theory: 60 Hours & Tutorial: 30 Hours
Course Objectives: To develop skill in students about i) Fourier series, Fourier and Laplace transforms as a tool to solve various problems of Mathematics. ii) Integral transforms to be used in the field of applied Mathematics and especially in the field of physics and electronics to express periodic functions that comprise communication signal in waveform. Course Outcome: After the completion of this course, students shall be able to use Fourier and Laplace transforms to solve the differential equations and to understand signal processing in frequency and time domain.	
Theory: 4 Credi	
Unit- I Fourier Series: Introduction, Periodic functions: Properties, Even & Odd functions. Special waveforms: Square wave, Sawtooth wave, and Triangular wave. Euler's Formulae for Fourier Series, Fourier Series for functions of period 2π, Fourier Series for functions of period $2L$, Dirichlet's conditions, Sum of Fourier series. If $f(x)$ is bounded and integrable function on $(-\pi, \pi)$ and if a_n, b_n are its Fourier coefficients, then $\sum a_n^2 + b_n^2$ converges. Half Range Series for sine and cosine functions, examples. Riemann Lebesque theorem.	
Unit – II The Fourier Transform. Periodic functions, Definition and examples of Fourier series, Dirichlet's conditions, determination of Fourier coefficients, even and odd functions and their Fourier expansion, change of interval, half range series. Fourier transform, inverse Fourier transform, Fourier sine and cosine transforms and their inversion, properties of Fourier transforms. Fourier transform of the derivative, convolution theorem, discrete Fourier transform and fast Fourier transform and their properties.	
Unit – III Definition, Laplace transform of elementary functions, Properties of Laplace transforms viz Linearity, translation, Change of Scale property etc. Laplace transform as periodic functions, Dirac-Delta function, Inverse Laplace transform. Laplace transform for derivatives, Laplace transform for integrals, Convolution theorem. Solution of ordinary differential equations with constant coefficients. Applications of partial differential equations.	
Unit – IV Application of Laplace transform to differential equation and integral equation. Application of Laplace transform to boundary value problems. Electrical circuits, dynamics, Beams, Heat conduction equations and wave equations.	
Tutorials: 2 Credits	
Unit – V Problems based on unit-I and unit-II.	
Unit – VI Problems based on unit-III and unit-IV.	
Recommended Books: 1. Ruel V.Churchill, Fourier Series & Boundary Value Problems, 8th Edition McGraw Hill Education 2011. 2. Davies, Brian, Integral Transforms and Their Applications, Springer, 2002. 3. Erwin, Kreysgiz, Advanced Engineering Mathematics, John Willey & Sons. 10th Edition, 2011. 4. K.S. Rao, Introduction to Partial Differential Equations, K.S. Rao, PHI, India. 5. Murrey R. Spiegel, Laplace Transforms, Schaum's outline series. 6. I. N. Sneddon: The use of Integral Transforms, McGraw-Hill, Singapore 1972. 7. R. R. Goldberg, Fourier Transforms, Cambridge University Press, 1961. 8. D. Brain, Integral Transforms and their applications, Springer, 2002.	



LAPACE AND FOURIER TRANSFORM (MMT522J3)



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Chapter 1

Fourier Series and Transforms

1.1 Introduction to Fourier Series

1.1.1 Periodic Functions

The central objects of study for Fourier series are periodic functions.

Definition 1.1.1 (Periodic Function). A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to be **periodic** with a **period** $T > 0$ if for every x in its domain,

$$f(x + T) = f(x)$$

The smallest positive value of T for which this relation holds is called the **fundamental period** of the function.

Example 1.1.2. 1. The function $f(x) = \sin(x)$ is periodic with periods $2\pi, 4\pi, 6\pi, \dots$. Its fundamental period is $T = 2\pi$, since this is the smallest positive value for which $\sin(x + T) = \sin(x)$.

2. Similarly, $g(x) = \cos(x)$ has a fundamental period of $T = 2\pi$.

3. The function $h(x) = \tan(x)$ has a fundamental period of $T = \pi$.

4. A constant function, $f(x) = c$, is periodic for any period $T > 0$. It does not have a fundamental period.

Properties of Periodic Functions

Let $f(x)$ and $g(x)$ be periodic functions with a common period T .

1. **Linear Combination:** The sum $h(x) = c_1 f(x) + c_2 g(x)$, where c_1 and c_2 are constants, is also periodic with period T .
2. **Product and Quotient:** The product $h(x) = f(x)g(x)$ and the quotient $q(x) = f(x)/g(x)$ (where $g(x) \neq 0$) are also periodic with period T .
3. **Horizontal Scaling:** If $f(x)$ has a period T , then the function $g(x) = f(kx)$ for some constant $k \neq 0$ is periodic with period $T/|k|$. For instance, since $\sin(x)$ has period 2π , the function $\sin(3x)$ has period $2\pi/3$.
4. **Least Common Multiple Period:** If $f(x)$ has period T_1 and $g(x)$ has period T_2 , and the ratio T_1/T_2 is a rational number, then the sum $f(x) + g(x)$ is periodic. Its fundamental period is the least common multiple (LCM) of T_1 and T_2 .

1.1.2 Even and Odd Functions

Recognizing the symmetry of a function can greatly simplify the calculation of its Fourier series.

Definition 1.1.3 (Even Function). A function $f(x)$ is **even** if for every x in its domain,

$$f(-x) = f(x)$$

Geometrically, the graph of an even function is symmetric with respect to the y-axis.

Examples: $x^2, x^4, \cos(x), |x|$.

Definition 1.1.4 (Odd Function). A function $f(x)$ is **odd** if for every x in its domain,

$$f(-x) = -f(x)$$

Geometrically, the graph of an odd function is symmetric with respect to the origin.

Examples: $x, x^3, \sin(x), \tan(x)$.

Properties of Even and Odd Functions

These properties are essential for simplifying Fourier coefficient calculations.

1. Algebraic Combinations:

- (Even) $\times$ (Even) = Even
- (Odd) $\times$ (Odd) = Even
- (Even) $\times$ (Odd) = Odd
- (Even) $\pm$ (Even) = Even
- (Odd) $\pm$ (Odd) = Odd

2. **Decomposition:** Any function $f(x)$ can be expressed as the sum of an even function and an odd function:

$$f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{Even Part}} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{Odd Part}}$$

3. **Integration over Symmetric Intervals:** Let $[-L, L]$ be an interval symmetric about the origin.

- If $f(x)$ is an **odd** function, then $\int_{-L}^L f(x) dx = 0$.
- If $f(x)$ is an **even** function, then $\int_{-L}^L f(x) dx = 2 \int_0^L f(x) dx$.

1.1.3 Special Periodic Waveforms

In engineering and physics, several fundamental periodic functions serve as building blocks for more complex signals. We define three of the most common ones here. For simplicity, we will define them over an interval of length 2π and state their fundamental period.

Square Wave

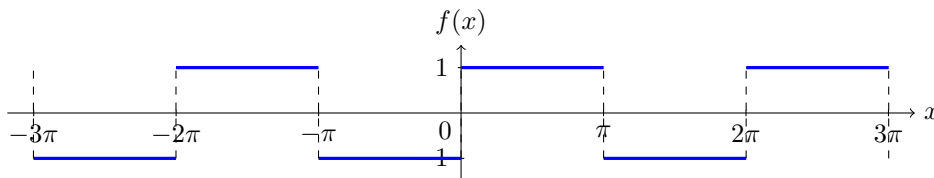
A square wave is a non-sinusoidal periodic waveform that alternates between two fixed voltage or current levels.

Definition 1.1.5. A standard square wave with period $T = 2\pi$ can be defined over the interval $(-\pi, \pi)$ as:

$$f(x) = \begin{cases} -1 & \text{if } -\pi < x < 0 \\ 1 & \text{if } 0 < x < \pi \end{cases}$$

and is extended periodically by the relation $f(x + 2\pi) = f(x)$. The function is typically left undefined at the points of discontinuity ($x = 0, \pm\pi, \pm2\pi, \dots$).

Symmetry: The square wave as defined above is an **odd function**, since $f(-x) = -f(x)$.



Sawtooth Wave

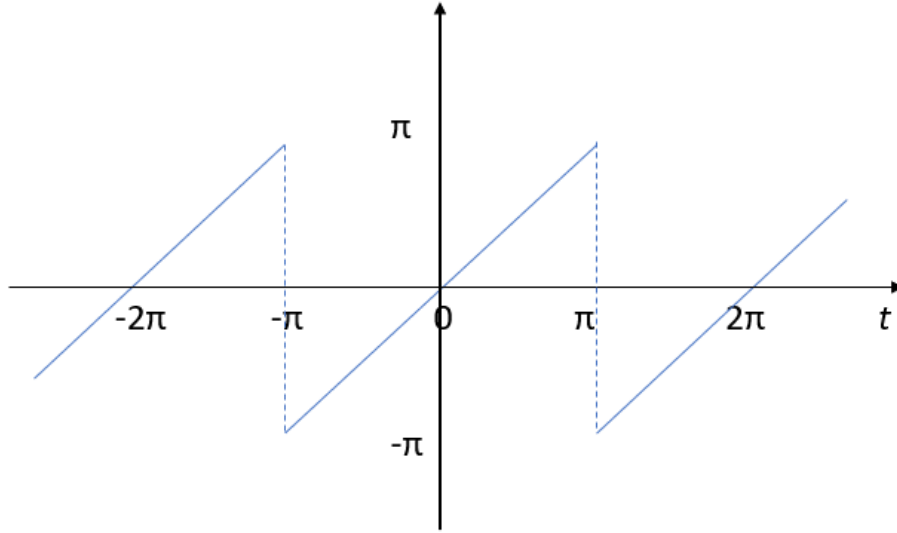
A sawtooth wave is a waveform that ramps up linearly and then sharply falls to its starting value, or vice-versa. It resembles the teeth of a saw.

Definition 1.1.6. A standard sawtooth wave with period $T = 2\pi$ can be defined over the interval $(-\pi, \pi)$ as:

$$f(x) = x$$

and is extended periodically by the relation $f(x + 2\pi) = f(x)$.

Symmetry: The sawtooth wave as defined above is an **odd function**.



Triangular Wave

A triangular wave is another non-sinusoidal waveform characterized by a linear rise and fall, creating a triangular shape.

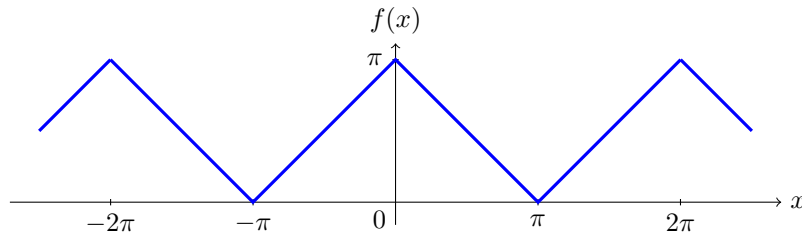
Definition 1.1.7. A standard triangular wave with period $T = 2\pi$ can be defined on the interval $[-\pi, \pi]$ using the absolute value function:

$$f(x) = \pi - |x|$$

and is extended periodically by the relation $f(x + 2\pi) = f(x)$. This can also be written piecewise as:

$$f(x) = \begin{cases} \pi + x & \text{if } -\pi \leq x < 0 \\ \pi - x & \text{if } 0 \leq x \leq \pi \end{cases}$$

Symmetry: The triangular wave as defined above is an **even function**, since $f(-x) = \pi - |-x| = \pi - |x| = f(x)$.



1.2 The Fourier Series Representation

The central goal of Fourier analysis is to represent a periodic function $f(x)$ as an infinite series of sine and cosine functions. This trigonometric series is known as the Fourier series.

1.2.1 Fourier Series for Functions of Period $2L$

Let us consider a function $f(x)$ that is periodic with a general period $T = 2L$. The Fourier series representation of $f(x)$ on an interval of length $2L$, for example $[-L, L]$, is given by:

Definition 1.2.1 (Fourier Series). The Fourier series of a function $f(x)$ defined on the interval $[-L, L]$ is given by

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right)$$

where the coefficients a_0 , a_n , and b_n are the **Fourier coefficients**. The symbol $\sim$ is used to indicate that the series on the right corresponds to the function $f(x)$, without making any claims about convergence at this stage.

Remark 1.2.2. The term $\frac{a_0}{2}$ is the average value (or DC component) of the function over one period. The reason for the factor of $1/2$ is for notational convenience, as it allows the general formula for a_n to also work for $n = 0$.

1.2.2 Euler's Formulae for the Fourier Coefficients

The Fourier coefficients are determined by a set of integral formulas known as Euler's formulae. These are derived by exploiting the **orthogonality properties** of sine and cosine functions over the interval $[-L, L]$. The key integral relations are:

$$\begin{aligned} \bullet \int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right) dx &= \begin{cases} 0 & \text{if } m \neq n \\ L & \text{if } m = n \neq 0 \\ 2L & \text{if } m = n = 0 \end{cases} \\ \bullet \int_{-L}^L \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx &= \begin{cases} 0 & \text{if } m \neq n \\ L & \text{if } m = n \neq 0 \end{cases} \\ \bullet \int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx &= 0 \text{ for all integers } m, n. \end{aligned}$$

By integrating the Fourier series term-by-term (a process that can be rigorously justified), we arrive at Euler's formulae.

Calculating the Coefficients

- **To find a_0 :** Integrate the Fourier series from $-L$ to L . Due to orthogonality, every sine and cosine term integrates to zero.

$$\int_{-L}^L f(x) dx = \int_{-L}^L \frac{a_0}{2} dx = \frac{a_0}{2} (2L) = a_0 L$$

- **To find a_n ($n \geq 1$):** Multiply the entire series by $\cos\left(\frac{n\pi x}{L}\right)$ and integrate from $-L$ to L . All terms integrate to zero except for the one where $n = m$.

$$\int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx = a_n \int_{-L}^L \cos^2\left(\frac{n\pi x}{L}\right) dx = a_n L$$

- **To find b_n ($n \geq 1$):** Multiply the series by $\sin\left(\frac{n\pi x}{L}\right)$ and integrate. All terms integrate to zero except for the one where $n = m$.

$$\int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx = b_n \int_{-L}^L \sin^2\left(\frac{n\pi x}{L}\right) dx = b_n L$$

This leads to the definitive formulas:

Theorem 1.2.3 (Euler's Formulae). *For a function $f(x)$ with period $2L$, the Fourier coefficients are given by:*

$$\begin{aligned} a_0 &= \frac{1}{L} \int_{-L}^L f(x) dx \\ a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \quad (n = 1, 2, 3, \dots) \\ b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \quad (n = 1, 2, 3, \dots) \end{aligned}$$

1.2.3 Fourier Series for Functions of Period 2π

A very common and important case is when the period of the function is $T = 2\pi$. This corresponds to setting $L = \pi$ in the general formulas. The term $\frac{n\pi x}{L}$ simplifies to $\frac{n\pi x}{\pi} = nx$.

The Fourier series for a function $f(x)$ with period 2π on the interval $[-\pi, \pi]$ is:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

The corresponding Euler's formulae for the coefficients become:

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \\ a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx \quad (n = 1, 2, 3, \dots) \\ b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx \quad (n = 1, 2, 3, \dots) \end{aligned}$$

Remark 1.2.4 (Interval of Integration). The interval of integration must have a length equal to the period $T = 2L$. While $[-L, L]$ is common, any interval such as $[0, 2L]$ or $[c, c + 2L]$ can be used. For a period of 2π , the interval $[0, 2\pi]$ is frequently used instead of $[-\pi, \pi]$. The choice often depends on the definition of the function.

Example 1.2.5 (Fourier Series of a Square Wave). Find the Fourier series for the square wave function $f(x)$ with period $T = 2\pi$ defined on $(-\pi, \pi)$ as:

$$f(x) = \begin{cases} -1 & \text{if } -\pi < x < 0 \\ 1 & \text{if } 0 < x < \pi \end{cases}$$

Solution. The function is periodic with period 2π , so we set $L = \pi$. By inspection, $f(-x) = -f(x)$, so the function is odd. Consequently, all cosine coefficients are zero: $a_n = 0$ for $n \geq 0$. We need only compute the sine coefficients, b_n . Since $f(x) \sin(nx)$ is an even function (a product of two odd functions), we can write:

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx = \frac{2}{\pi} \int_0^{\pi} (1) \sin(nx) dx \\ &= \frac{2}{\pi} \left[-\frac{\cos(nx)}{n} \right]_0^{\pi} = \frac{2}{n\pi} (1 - \cos(n\pi)) \end{aligned}$$

The term $(1 - \cos(n\pi))$ is 0 if n is even, and 2 if n is odd. Thus, $b_n = 0$ for even n and $b_n = \frac{4}{n\pi}$ for odd n . The resulting Fourier series is composed only of sine terms with odd harmonics:

$$f(x) \sim \sum_{n=1,3,5,\dots}^{\infty} \frac{4}{n\pi} \sin(nx) = \frac{4}{\pi} \left(\sin(x) + \frac{1}{3} \sin(3x) + \frac{1}{5} \sin(5x) + \dots \right)$$

■

Example 1.2.6 (Fourier Series of a Triangular Wave). Find the Fourier series for the triangular wave function $f(x) = \pi - |x|$ with period $T = 2\pi$ on the interval $[-\pi, \pi]$.

Solution. The function has period 2π ($L = \pi$) and is an even function since $f(-x) = \pi - |-x| = \pi - |x| = f(x)$. Therefore, all sine coefficients are zero: $b_n = 0$ for all $n \geq 1$. The DC component, a_0 , is calculated as:

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} (\pi - |x|) dx = \frac{2}{\pi} \int_0^{\pi} (\pi - x) dx = \frac{2}{\pi} \left[\pi x - \frac{x^2}{2} \right]_0^{\pi} = \pi$$

For the cosine coefficients a_n ($n \geq 1$), the integrand $(\pi - |x|) \cos(nx)$ is even:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} (\pi - |x|) \cos(nx) dx = \frac{2}{\pi} \int_0^{\pi} (\pi - x) \cos(nx) dx$$

Using integration by parts, we find:

$$\begin{aligned} a_n &= \frac{2}{\pi} \left(\left[(\pi - x) \frac{\sin(nx)}{n} \right]_0^{\pi} + \frac{1}{n} \int_0^{\pi} \sin(nx) dx \right) \\ &= \frac{2}{n\pi} \left[-\frac{\cos(nx)}{n} \right]_0^{\pi} = \frac{2}{n^2\pi} (1 - \cos(n\pi)) = \frac{2}{n^2\pi} (1 - (-1)^n) \end{aligned}$$

This yields $a_n = 0$ for even n and $a_n = \frac{4}{n^2\pi}$ for odd n . The Fourier series is:

$$\begin{aligned} f(x) &\sim \frac{a_0}{2} + \sum_{n=1,3,5,\dots}^{\infty} a_n \cos(nx) = \frac{\pi}{2} + \frac{4}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{\cos(nx)}{n^2} \\ f(x) &\sim \frac{\pi}{2} + \frac{4}{\pi} \left(\cos(x) + \frac{1}{9} \cos(3x) + \frac{1}{25} \cos(5x) + \dots \right) \end{aligned}$$

■

Example 1.2.7 (Fourier Series for Period $2L$). Find the Fourier series for the function $f(x) = x$ defined on the interval $[0, 2)$ with period $T = 2$.

Solution. The function has period $T = 2$, so $2L = 2$ which implies $L = 1$. We compute the coefficients by integrating over the interval $[0, 2]$. The DC component a_0 is:

$$a_0 = \frac{1}{1} \int_0^2 x dx = \left[\frac{x^2}{2} \right]_0^2 = 2$$

The cosine coefficients a_n for $n \geq 1$ are:

$$a_n = \int_0^2 x \cos(n\pi x) dx = \left[\frac{x \sin(n\pi x)}{n\pi} \right]_0^2 - \int_0^2 \frac{\sin(n\pi x)}{n\pi} dx = 0 - \left[\frac{\cos(n\pi x)}{n^2\pi^2} \right]_0^2 = 0$$

The sine coefficients b_n for $n \geq 1$ are:

$$\begin{aligned} b_n &= \int_0^2 x \sin(n\pi x) dx = \left[\frac{-x \cos(n\pi x)}{n\pi} \right]_0^2 + \int_0^2 \frac{\cos(n\pi x)}{n\pi} dx \\ &= -\frac{2 \cos(2n\pi)}{n\pi} + \left[\frac{\sin(n\pi x)}{n^2\pi^2} \right]_0^2 = -\frac{2}{n\pi} \end{aligned}$$

Assembling the series with $a_0 = 2$, $a_n = 0$, and $b_n = -2/(n\pi)$:

$$\begin{aligned} f(x) &\sim \frac{a_0}{2} + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) = \frac{2}{2} + \sum_{n=1}^{\infty} \left(-\frac{2}{n\pi}\right) \sin(n\pi x) \\ f(x) &\sim 1 - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{\sin(n\pi x)}{n} = 1 - \frac{2}{\pi} \left(\sin(\pi x) + \frac{1}{2} \sin(2\pi x) + \frac{1}{3} \sin(3\pi x) + \dots \right) \end{aligned}$$

■

Exercise 1.2.8. Find the Fourier series expansion for each of the following functions:

1. Find the Fourier series of the square wave function defined by:

$$f(x) = \begin{cases} \pi + x & \text{for } -\pi < x < 0 \\ 0 & \text{for } 0 \leq x < \pi, \end{cases} \quad f(x + 2\pi) = f(x).$$

with period 2π .

2. Determine the Fourier series of the sawtooth wave function:

$$f(x) = \begin{cases} 2 + x & \text{for } -2 \leq x \leq 0 \\ 2 - x & \text{for } 0 \leq x \leq 2, \end{cases} \quad f(x + 4) = f(x).$$

with period 2π .

3. Compute the Fourier series of the triangular wave function:

$$f(x) = \begin{cases} \pi + x & \text{for } -\pi \leq x \leq 0 \\ \pi - x & \text{for } 0 \leq x \leq \pi \end{cases}$$

with period 2π .

For each function:

- Calculate the Fourier coefficients a_0 , a_n , and b_n
- Write the complete Fourier series expansion
- State the value to which the series converges at points of discontinuity (if any)

1.3 Fourier Series of Even and Odd Functions

Definition 1.3.1 (Even Function). A function $f(x)$ is **even** if for all x in its domain:

$$f(-x) = f(x)$$

The graph of an even function is symmetric about the y-axis.

Definition 1.3.2 (Odd Function). A function $f(x)$ is **odd** if for all x in its domain:

$$f(-x) = -f(x)$$

The graph of an odd function is symmetric about the origin.

Fourier Series Simplifications

For a function $f(x)$ with period $2L$:

- If $f(x)$ is **even**, then all b_n coefficients vanish:

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx = 0$$

The Fourier series becomes a **Fourier cosine series**:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

where

$$a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

- If $f(x)$ is **odd**, then all a_n coefficients vanish:

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx = 0$$

The Fourier series becomes a **Fourier sine series**:

$$f(x) \sim \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

where

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

Example 1.3.3 (Even Function). Find the Fourier series of $f(x) = x^2$ on $[-\pi, \pi]$.

Solution: Since $f(-x) = (-x)^2 = x^2 = f(x)$, the function is even. We compute the coefficients:

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x^2 dx = \frac{2}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi} = \frac{2\pi^2}{3}$$

For $n \geq 1$:

$$a_n = \frac{2}{\pi} \int_0^{\pi} x^2 \cos(nx) dx$$

Using integration by parts twice:

$$a_n = \frac{2}{\pi} \left[\frac{x^2 \sin(nx)}{n} + \frac{2x \cos(nx)}{n^2} - \frac{2 \sin(nx)}{n^3} \right]_0^{\pi} = \frac{2}{\pi} \left(\frac{2\pi \cos(n\pi)}{n^2} \right) = \frac{4(-1)^n}{n^2}$$

Thus, the Fourier series is:

$$x^2 \sim \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2} \cos(nx)$$

Example 1.3.4 (Odd Function). Find the Fourier series of $f(x) = x$ on $[-\pi, \pi]$.

Solution: Since $f(-x) = -x = -f(x)$, the function is odd. We compute the coefficients:

$$b_n = \frac{2}{\pi} \int_0^{\pi} x \sin(nx) dx$$

Using integration by parts:

$$b_n = \frac{2}{\pi} \left[-\frac{x \cos(nx)}{n} + \frac{\sin(nx)}{n^2} \right]_0^{\pi} = \frac{2}{\pi} \left(-\frac{\pi \cos(n\pi)}{n} \right) = \frac{2(-1)^{n+1}}{n}$$

Thus, the Fourier series is:

$$x \sim \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n} \sin(nx)$$

Example 1.3.5 (Mixed Function). Find the Fourier series of $f(x) = \begin{cases} x & \text{for } 0 \leq x \leq \pi \\ 0 & \text{for } -\pi \leq x < 0 \end{cases}$ on $[-\pi, \pi]$.

Solution: This function is neither even nor odd. We compute all coefficients:

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} x dx = \frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} = \frac{\pi}{2}$$

For $n \geq 1$:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx = \frac{1}{\pi} \int_0^{\pi} x \cos(nx) dx$$

Using integration by parts:

$$a_n = \frac{1}{\pi} \left[\frac{x \sin(nx)}{n} + \frac{\cos(nx)}{n^2} \right]_0^{\pi} = \frac{1}{\pi} \left(\frac{\cos(n\pi)}{n^2} - \frac{1}{n^2} \right) = \frac{(-1)^n - 1}{\pi n^2}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx = \frac{1}{\pi} \int_0^{\pi} x \sin(nx) dx$$

Using integration by parts:

$$b_n = \frac{1}{\pi} \left[-\frac{x \cos(nx)}{n} + \frac{\sin(nx)}{n^2} \right]_0^{\pi} = \frac{1}{\pi} \left(-\frac{\pi \cos(n\pi)}{n} \right) = \frac{(-1)^{n+1}}{n}$$

Thus, the Fourier series is:

$$f(x) \sim \frac{\pi}{4} + \sum_{n=1}^{\infty} \left[\frac{(-1)^n - 1}{\pi n^2} \cos(nx) + \frac{(-1)^{n+1}}{n} \sin(nx) \right]$$

1.4 Convergence of Fourier Series

So far, we have used the symbol $\sim$ to relate a function to its Fourier series. This was a deliberate choice because we have not yet established the conditions under which the infinite series actually converges, and if it does, whether it converges to the original function $f(x)$. The following sections address this crucial question.

1.4.1 Dirichlet's Conditions

The conditions that guarantee the convergence of a Fourier series are known as Dirichlet's conditions. Most periodic functions encountered in science and engineering satisfy these conditions.

Definition 1.4.1 (Dirichlet's Conditions). A function $f(x)$ defined on the interval $[-L, L]$ is said to satisfy **Dirichlet's conditions** if it meets the following criteria:

1. $f(x)$ is single-valued and periodic with period $2L$.
2. $f(x)$ has a finite number of finite discontinuities within one period. (The function is **piecewise continuous**).
3. $f(x)$ has a finite number of maxima and minima within one period. (The function is of **bounded variation**, or more simply, **piecewise monotonic**).

Remark 1.4.2. In simpler terms, these conditions mean that the function must be "well-behaved". It cannot have an infinite number of wiggles or jumps in a single period, nor can it shoot off to infinity. The square wave, sawtooth wave, and triangular wave all satisfy Dirichlet's conditions.

Riemann-Lebesgue Lemma

Theorem 1.4.3 (Riemann-Lebesgue Lemma). Let f be an absolutely integrable function on the interval $[a, b]$, i.e.,

$$\int_a^b |f(t)| dt < \infty.$$

Then:

$$\lim_{n \rightarrow \infty} \int_a^b f(t) \sin(nt) dt = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \int_a^b f(t) \cos(nt) dt = 0.$$

In particular, for Fourier coefficients of an integrable periodic function, we have:

$$\lim_{n \rightarrow \infty} a_n = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} b_n = 0.$$

Proof. We prove the result for the sine integral; the cosine case is analogous.

First consider a step function $\phi(t)$ on $[a, b]$, which is constant on subintervals:

$$\phi(t) = c_j \quad \text{for } t \in (t_{j-1}, t_j), \quad j = 1, \dots, m.$$

Then:

$$\int_a^b \phi(t) \sin(nt) dt = \sum_{j=1}^m c_j \int_{t_{j-1}}^{t_j} \sin(nt) dt = \sum_{j=1}^m \frac{c_j}{n} [-\cos(nt)]_{t_{j-1}}^{t_j}.$$

Since $|\cos(nt)| \leq 1$, we have:

$$\left| \int_a^b \phi(t) \sin(nt) dt \right| \leq \sum_{j=1}^m \frac{2|c_j|}{n} \leq \frac{2M}{n} \sum_{j=1}^m (t_j - t_{j-1}) = \frac{2M(b-a)}{n},$$

where $M = \max_j |c_j|$. Clearly, this tends to 0 as $n \rightarrow \infty$.

For any integrable function f and any $\epsilon > 0$, there exists a step function ϕ such that:

$$\int_a^b |f(t) - \phi(t)| dt < \frac{\epsilon}{2}.$$

Then:

$$\left| \int_a^b f(t) \sin(nt) dt \right| \leq \left| \int_a^b [f(t) - \phi(t)] \sin(nt) dt \right| + \left| \int_a^b \phi(t) \sin(nt) dt \right|.$$

The first term is bounded by:

$$\int_a^b |f(t) - \phi(t)| dt < \frac{\epsilon}{2}.$$

For the second term, since ϕ is a step function, by Step 1 there exists n_0 such that for all $n > n_0$:

$$\left| \int_a^b \phi(t) \sin(nt) dt \right| < \frac{\epsilon}{2}.$$

Hence, for $n > n_0$:

$$\left| \int_a^b f(t) \sin(nt) dt \right| < \epsilon,$$

which proves the result.

The application to Fourier coefficients follows by taking:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt) dt, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt,$$

and applying the lemma with $n = n$. ■

1.4.2 The Sum of the Fourier Series (Convergence Theorem)

If a function satisfies Dirichlet's conditions, its Fourier series is guaranteed to converge. The value to which it converges is given by the following theorem.

Theorem 1.4.4 (Fourier Convergence Theorem). *Let $f(x)$ be a periodic function with period $2L$ that satisfies Dirichlet's conditions. The Fourier series of $f(x)$ converges:*

1. to $f(x)$ at all points where $f(x)$ is continuous.
2. to the average of the left-hand and right-hand limits at any point of discontinuity. That is, at a point of discontinuity x_0 , the series converges to:

$$\frac{f(x_0^+) + f(x_0^-)}{2}$$

where $f(x_0^+) = \lim_{h \rightarrow 0^+} f(x_0 + h)$ and $f(x_0^-) = \lim_{h \rightarrow 0^+} f(x_0 - h)$.

Proof. We present the proof for a function with period 2π ; the case for period $2L$ follows by a change of variable.

Let $S_N(x)$ be the N -th partial sum of the Fourier series of f :

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^N (a_n \cos(nx) + b_n \sin(nx)).$$

Substituting the expressions for the Fourier coefficients:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt) dt, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt,$$

we obtain:

$$S_N(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \left[\frac{1}{2} + \sum_{n=1}^N \cos(n(t-x)) \right] dt.$$

The expression in brackets is the Dirichlet kernel:

$$D_N(u) = \frac{1}{2} + \sum_{n=1}^N \cos(nu) = \frac{\sin((N + \frac{1}{2})u)}{2 \sin(u/2)}.$$

Thus,

$$S_N(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) D_N(t-x) dt.$$

Making the substitution $u = t - x$, and using the periodicity of the functions, we get:

$$S_N(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x+u) D_N(u) du = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x+u) \frac{\sin((N + \frac{1}{2})u)}{\sin(u/2)} du.$$

Since the integrand is even in u , we may write:

$$S_N(x) = \frac{1}{2\pi} \int_0^{\pi} [f(x+u) + f(x-u)] \frac{\sin((N + \frac{1}{2})u)}{\sin(u/2)} du.$$

Noting that $\int_0^{\pi} \frac{\sin kx}{x} = \pi$, we consider the difference:

$$S_N(x) - \frac{f(x^+) + f(x^-)}{2} = \frac{1}{2\pi} \int_0^{\pi} \phi(u) \frac{\sin((N + \frac{1}{2})u)}{\sin(u/2)} du,$$

where $\phi(u) = f(x+u) + f(x-u) - (f(x^+) + f(x^-))$.

Since f satisfies Dirichlet's conditions, $\phi(u)/\sin(u/2)$ is absolutely integrable on $[0, \pi]$. By the Riemann–Lebesgue Lemma:

$$\lim_{N \rightarrow \infty} \int_0^{\pi} \phi(u) \frac{\sin((N + \frac{1}{2})u)}{\sin(u/2)} du = 0.$$

Therefore,

$$\lim_{N \rightarrow \infty} S_N(x) = \frac{f(x^+) + f(x^-)}{2}.$$

If f is continuous at x , then $f(x^+) = f(x^-) = f(x)$, and the series converges to $f(x)$. ■

Example 1.4.5 (Convergence of the Square Wave). Consider the square wave on $(-\pi, \pi)$ from the previous examples.

$$f(x) = \begin{cases} -1 & \text{if } -\pi < x < 0 \\ 1 & \text{if } 0 < x < \pi \end{cases}$$

The function is continuous everywhere except at $x = k\pi$ for integers k .

- At a point of continuity, like $x = \pi/2$, the Fourier series converges to $f(\pi/2) = 1$.
- At the discontinuity $x = 0$, the left-hand limit is $f(0^-) = -1$ and the right-hand limit is $f(0^+) = 1$. The series converges to the average:

$$\frac{f(0^+) + f(0^-)}{2} = \frac{1 + (-1)}{2} = 0$$

This is a remarkable result. The series, which consists of only sine terms (all of which are zero at $x = 0$), correctly converges to the midpoint of the jump.

Example 1.4.6. Consider the function $f(x) = x$ for $-\pi < x < \pi$.

1. Find the Fourier series for $f(x)$.
2. Discuss the convergence of the Fourier series at $x = \pi/2$, $x = \pi$, and $x = -\pi$.

Solution. 1. The function $f(x) = x$ is an odd function. Therefore, $a_n = 0$ for all n . We only need to calculate b_n . The period of the function is $2L = 2\pi$, so $L = \pi$.

$$\begin{aligned} b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin(nx) dx \end{aligned}$$

We integrate by parts with $u = x$, $dv = \sin(nx)dx$, so $du = dx$, $v = -\frac{1}{n} \cos(nx)$.

$$\begin{aligned} b_n &= \frac{1}{\pi} \left[-\frac{x}{n} \cos(nx) \right]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} -\frac{1}{n} \cos(nx) dx \\ &= \frac{1}{\pi} \left[\left(-\frac{\pi}{n} \cos(n\pi) \right) - \left(\frac{\pi}{n} \cos(-n\pi) \right) \right] + \frac{1}{n\pi} \int_{-\pi}^{\pi} \cos(nx) dx \\ &= \frac{1}{\pi} \left[-\frac{\pi}{n} (-1)^n - \frac{\pi}{n} (-1)^n \right] + \frac{1}{n\pi} \left[\frac{1}{n} \sin(nx) \right]_{-\pi}^{\pi} \\ &= \frac{1}{\pi} \left[-\frac{2\pi}{n} (-1)^n \right] + \frac{1}{n^2\pi} [\sin(n\pi) - \sin(-n\pi)] \\ &= -\frac{2}{n} (-1)^n + \frac{1}{n^2\pi} [0 - 0] \\ &= \frac{2}{n} (-1)^{n+1} \end{aligned}$$

Thus, the Fourier series for $f(x) = x$ is:

$$f(x) \sim \sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin(nx)$$

2. We use Dirichlet's Convergence Theorem. The function $f(x) = x$ is piecewise smooth on the interval $[-\pi, \pi]$. Its derivative $f'(x) = 1$ is continuous on $(-\pi, \pi)$.

(a) At $x = \pi/2$: Since $f(x)$ is continuous at $x = \pi/2$ (and in the open interval $(-\pi, \pi)$), the Fourier series converges to $f(\pi/2)$.

$$\sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin(n\pi/2) = f(\pi/2) = \pi/2$$

(b) At $x = \pi$: This is an endpoint of the interval. According to the convergence theorem, the Fourier series converges to the average of the left-hand and right-hand limits of the periodic extension of $f(x)$. The periodic extension $f_{ext}(x)$ at $x = \pi$ has a left-hand limit $f(\pi^-) = \pi$. The right-hand limit for the periodic extension at $x = \pi$ is $f(-\pi^+) = -\pi$.

$$\sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin(n\pi) = \frac{f(\pi^-) + f(-\pi^+)}{2} = \frac{\pi + (-\pi)}{2} = 0$$

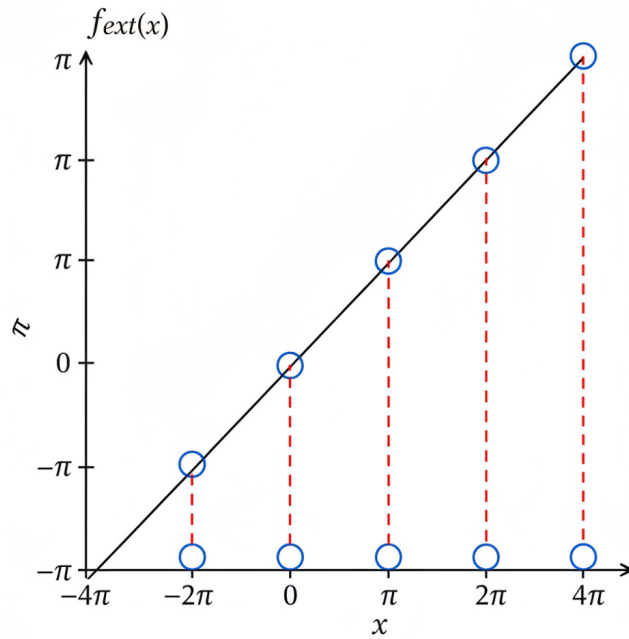
(Note that $\sin(n\pi) = 0$ for all integers n , so the sum is indeed 0).

(c) At $x = -\pi$: Similar to $x = \pi$, this is an endpoint. The left-hand limit for the periodic extension at $x = -\pi$ is $f(\pi^-) = \pi$. The right-hand limit at $x = -\pi$ is $f(-\pi^+) = -\pi$.

$$\sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin(-n\pi) = \frac{f(\pi^-) + f(-\pi^+)}{2} = \frac{\pi + (-\pi)}{2} = 0$$

(Again, $\sin(-n\pi) = 0$, so the sum is 0).

The graph of the function $f(x) = x$ and its periodic extension shows the jumps at integer multiples of π .



■

Example 1.4.7. Consider the function $f(x)$ defined as:

$$f(x) = \begin{cases} 0 & -\pi < x < 0 \\ 1 & 0 \leq x < \pi \end{cases}$$

1. Find the Fourier series for $f(x)$.
2. Discuss the convergence of the Fourier series at $x = -\pi/2$, $x = 0$, and $x = \pi$.

Solution. 1. The period of the function is $2L = 2\pi$, so $L = \pi$. First, calculate a_0 :

$$\begin{aligned} a_0 &= \frac{1}{L} \int_{-L}^L f(x) dx \\ &= \frac{1}{\pi} \left[\int_{-\pi}^0 0 dx + \int_0^{\pi} 1 dx \right] \\ &= \frac{1}{\pi} [0 + (x)_0^{\pi}] \\ &= \frac{1}{\pi} [\pi - 0] = 1 \end{aligned}$$

Next, calculate a_n :

$$\begin{aligned} a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \\ &= \frac{1}{\pi} \left[\int_{-\pi}^0 0 \cdot \cos(nx) dx + \int_0^{\pi} 1 \cdot \cos(nx) dx \right] \\ &= \frac{1}{\pi} \left[\frac{1}{n} \sin(nx) \right]_0^{\pi} \\ &= \frac{1}{\pi n} [\sin(n\pi) - \sin(0)] \\ &= \frac{1}{\pi n} [0 - 0] = 0 \end{aligned}$$

Now, calculate b_n :

$$\begin{aligned}
 b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \\
 &= \frac{1}{\pi} \left[\int_{-\pi}^0 0 \cdot \sin(nx) dx + \int_0^{\pi} 1 \cdot \sin(nx) dx \right] \\
 &= \frac{1}{\pi} \left[-\frac{1}{n} \cos(nx) \right]_0^{\pi} \\
 &= -\frac{1}{\pi n} [\cos(n\pi) - \cos(0)] \\
 &= -\frac{1}{\pi n} [(-1)^n - 1]
 \end{aligned}$$

If n is even, $(-1)^n = 1$, so $b_n = -\frac{1}{\pi n}[1 - 1] = 0$. If n is odd, $(-1)^n = -1$, so $b_n = -\frac{1}{\pi n}[-1 - 1] = \frac{2}{\pi n}$. So,
 $b_n = \begin{cases} \frac{2}{\pi n} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$. The Fourier series for $f(x)$ is:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

$$f(x) \sim \frac{1}{2} + \sum_{n \text{ odd}} \frac{2}{\pi n} \sin(nx)$$

We can write this by letting $n = 2k - 1$ for $k = 1, 2, \dots$:

$$f(x) \sim \frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{1}{2k-1} \sin((2k-1)x)$$

2. We use Dirichlet's Convergence Theorem. The function $f(x)$ is piecewise smooth. It has a jump discontinuity at $x = 0$. The derivative is $f'(x) = 0$ for $x \in (-\pi, 0)$ and $x \in (0, \pi)$.

- (a) At $x = -\pi/2$: The function is continuous at $x = -\pi/2$, and $f(-\pi/2) = 0$.

$$\frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{1}{2k-1} \sin((2k-1)(-\pi/2)) = f(-\pi/2) = 0$$

- (b) At $x = 0$: This is a point of discontinuity. The Fourier series converges to the average of the left-hand and right-hand limits. $f(0^-) = \lim_{x \rightarrow 0^-} f(x) = 0$. $f(0^+) = \lim_{x \rightarrow 0^+} f(x) = 1$.

$$\frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{1}{2k-1} \sin(0) = \frac{f(0^-) + f(0^+)}{2} = \frac{0+1}{2} = \frac{1}{2}$$

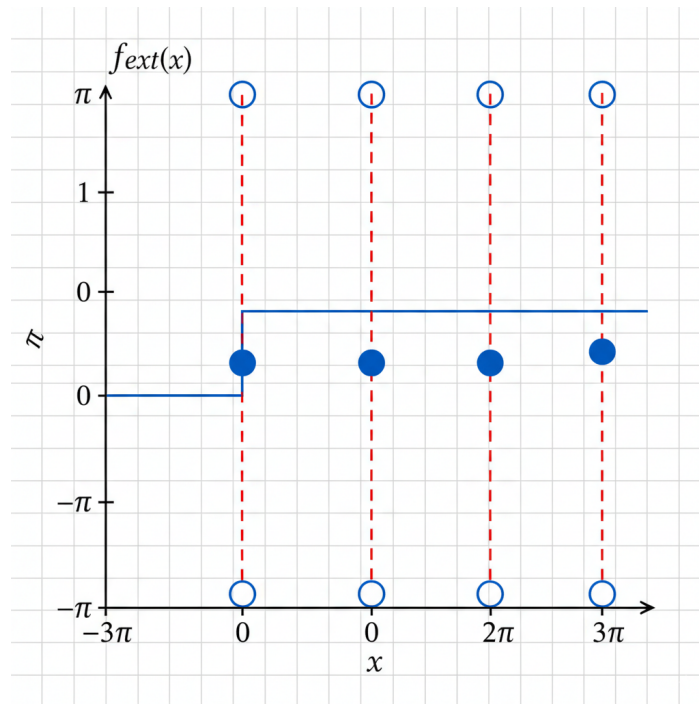
(Note that $\sin(0) = 0$, so the sum evaluates to $1/2$, which is correct).

- (c) At $x = \pi$: This is an endpoint and also a point of discontinuity in the periodic extension. The left-hand limit is $f(\pi^-) = 1$. The right-hand limit for the periodic extension at $x = \pi$ is $f(-\pi^+) = 0$.

$$\frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{1}{2k-1} \sin((2k-1)\pi) = \frac{f(\pi^-) + f(-\pi^+)}{2} = \frac{1+0}{2} = \frac{1}{2}$$

(Note that $\sin((2k-1)\pi) = 0$ for all integers k , so the sum evaluates to $1/2$, which is correct).

The graph of the function $f(x)$ and its periodic extension, showing the jumps at integer multiples of π .



■

Fourier series for sum of infinite series

Fourier series expansions can be used to evaluate the sums of certain infinite series by choosing appropriate values for x in the series representation. Below are two classic examples demonstrating this technique.

Example 1.4.8. We wish to evaluate the series:

$$S_1 = \sum_{k=1}^{\infty} \frac{1}{(2k-1)^2} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots$$

Consider the even function $f(x) = |x|$ on the interval $[-\pi, \pi]$, extended periodically with period 2π . Its Fourier series expansion is:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx),$$

where the coefficients are:

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x dx = \pi, \quad a_n = \frac{2}{\pi} \int_0^{\pi} x \cos(nx) dx.$$

For $n \geq 1$, integration by parts yields:

$$a_n = \frac{2}{\pi} \left[\frac{x \sin(nx)}{n} + \frac{\cos(nx)}{n^2} \right]_0^{\pi} = \frac{2}{\pi} \cdot \frac{\cos(n\pi) - 1}{n^2} = \frac{2((-1)^n - 1)}{\pi n^2}.$$

Note that $a_n = 0$ for even n , and for odd $n = 2k - 1$:

$$a_{2k-1} = \frac{2((-1)^{2k-1} - 1)}{\pi(2k-1)^2} = -\frac{4}{\pi(2k-1)^2}.$$

Thus, the Fourier series becomes:

$$|x| = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\cos((2k-1)x)}{(2k-1)^2}.$$

Setting $x = 0$ (where $f(0) = 0$):

$$0 = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{1}{(2k-1)^2}.$$

Solving for the series:

$$\sum_{k=1}^{\infty} \frac{1}{(2k-1)^2} = \frac{\pi^2}{8}.$$

Example 1.4.9. We wish to evaluate the alternating series:

$$S_2 = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{2k-1} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$$

Consider the odd function $f(x) = x$ on the interval $(-\pi, \pi)$, extended periodically with period 2π . Its Fourier sine series expansion is:

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(nx),$$

where the coefficients are:

$$b_n = \frac{2}{\pi} \int_0^{\pi} x \sin(nx) dx.$$

Integration by parts yields:

$$b_n = \frac{2}{\pi} \left[-\frac{x \cos(nx)}{n} + \frac{\sin(nx)}{n^2} \right]_0^{\pi} = \frac{2}{\pi} \cdot \left(-\frac{\pi \cos(n\pi)}{n} \right) = \frac{2(-1)^{n+1}}{n}.$$

Thus, the Fourier series becomes:

$$x = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(nx).$$

Setting $x = \frac{\pi}{2}$ (where $f(\pi/2) = \pi/2$):

$$\frac{\pi}{2} = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin\left(\frac{n\pi}{2}\right).$$

Note that $\sin(n\pi/2)$ is nonzero only for odd n , with:

$$\sin\left(\frac{n\pi}{2}\right) = \begin{cases} 1 & \text{if } n \equiv 1 \pmod{4}, \\ -1 & \text{if } n \equiv 3 \pmod{4}. \end{cases}$$

Thus, the series becomes:

$$\frac{\pi}{2} = 2 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots \right).$$

Solving for the series:

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}.$$

Exercise 1.4.10. Use Fourier series to verify the following sums:

1. Using the Fourier series expansion of $f(x) = x^2$ on $[-\pi, \pi]$, show that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

2. Using the Fourier series expansion of $f(x) = x^2$ on $[-\pi, \pi]$, show that

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}.$$

3. Using the Fourier series expansion of $f(x) = x^4$ on $[-\pi, \pi]$, show that

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}.$$

4. Using the Fourier series expansion of $f(x) = |x|$ on $[-\pi, \pi]$, show that

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}.$$

Note: For each problem:

- Find the complete Fourier series expansion of the given function
- Choose an appropriate value of x to evaluate the series
- Justify why the Fourier series converges to the function value at your chosen point
- Manipulate the resulting equation to obtain the desired sum

1.5 Half-Range Fourier Series Expansions

When a function $f(x)$ is defined only on the interval $[0, L]$, we can extend it to the full interval $[-L, L]$ in two natural ways:

1. **Even extension:** Define $f_e(x) = f(-x)$ for $-L \leq x < 0$
2. **Odd extension:** Define $f_o(x) = -f(-x)$ for $-L \leq x < 0$

These extensions lead to Fourier series containing only cosine terms (even extension) or only sine terms (odd extension), known as **half-range expansions**.

1.6 Fourier Cosine Series

For a function $f(x)$ defined on $[0, L]$, its Fourier cosine series is given by:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

where the coefficients are:

$$a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx, \quad n = 0, 1, 2, \dots$$

This series represents the even periodic extension of $f(x)$ with period $2L$.

1.7 Fourier Sine Series

For a function $f(x)$ defined on $[0, L]$, its Fourier sine series is given by:

$$f(x) \sim \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

where the coefficients are:

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx, \quad n = 1, 2, 3, \dots$$

This series represents the odd periodic extension of $f(x)$ with period $2L$.

Example 1.7.1 (Fourier Cosine Series). Find the Fourier cosine series for $f(x) = x$ on $[0, \pi]$.

Solution: We have $L = \pi$. Compute the coefficients:

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x dx = \frac{2}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} = \pi$$

For $n \geq 1$:

$$a_n = \frac{2}{\pi} \int_0^{\pi} x \cos(nx) dx$$

Using integration by parts:

$$a_n = \frac{2}{\pi} \left[\frac{x \sin(nx)}{n} + \frac{\cos(nx)}{n^2} \right]_0^{\pi} = \frac{2}{\pi} \left(\frac{\cos(n\pi)}{n^2} - \frac{1}{n^2} \right) = \frac{2[(-1)^n - 1]}{\pi n^2}$$

Thus, the Fourier cosine series is:

$$x \sim \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\cos((2k-1)x)}{(2k-1)^2}$$

Example 1.7.2 (Fourier Sine Series). Find the Fourier sine series for $f(x) = 1$ on $[0, \pi]$.

Solution: We have $L = \pi$. Compute the coefficients:

$$b_n = \frac{2}{\pi} \int_0^{\pi} \sin(nx) dx = \frac{2}{\pi} \left[-\frac{\cos(nx)}{n} \right]_0^{\pi} = \frac{2[1 - (-1)^n]}{\pi n}$$

Thus, the Fourier sine series is:

$$1 \sim \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\sin((2k-1)x)}{2k-1}$$

Example 1.7.3. Find both the cosine and sine series for $f(x) = \pi - x$ on $[0, \pi]$.

Solution for cosine series:

$$a_0 = \frac{2}{\pi} \int_0^\pi (\pi - x) dx = \frac{2}{\pi} \left[\pi x - \frac{x^2}{2} \right]_0^\pi = \pi$$

For $n \geq 1$:

$$a_n = \frac{2}{\pi} \int_0^\pi (\pi - x) \cos(nx) dx = \frac{2}{\pi} \left[\frac{(\pi - x) \sin(nx)}{n} - \frac{\cos(nx)}{n^2} \right]_0^\pi = \frac{2[1 - (-1)^n]}{\pi n^2}$$

Thus, the cosine series is:

$$\pi - x \sim \frac{\pi}{2} + \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\cos((2k-1)x)}{(2k-1)^2}$$

Solution for sine series:

$$b_n = \frac{2}{\pi} \int_0^\pi (\pi - x) \sin(nx) dx = \frac{2}{\pi} \left[-\frac{(\pi - x) \cos(nx)}{n} - \frac{\sin(nx)}{n^2} \right]_0^\pi = \frac{2}{n}$$

Thus, the sine series is:

$$\pi - x \sim 2 \sum_{n=1}^{\infty} \frac{\sin(nx)}{n}$$

1.8 Applications

Half-range expansions are particularly useful in:

- Solving boundary value problems in partial differential equations
- Signal processing where only half-wave symmetry is present
- Problems with specific boundary conditions (e.g., insulated ends)

Exercise 1.8.1. Find the half-range Fourier expansions for the following functions defined on the interval $[0, \pi]$:

1. Find the Fourier cosine series for the function:

$$f(x) = x(\pi - x)$$

2. Find the Fourier sine series for the function:

$$f(x) = \begin{cases} x & \text{for } 0 \leq x \leq \frac{\pi}{2} \\ \pi - x & \text{for } \frac{\pi}{2} < x \leq \pi \end{cases}$$

3. For the function $f(x) = e^x$ on $[0, \pi]$:

- (a) Find the Fourier cosine series expansion
- (b) Find the Fourier sine series expansion

For each problem:

- Compute the appropriate Fourier coefficients (a_n for cosine series, b_n for sine series)
- Write the complete Fourier series expansion
- State the value to which the series converges at points of discontinuity (if any)
- Sketch the periodic extension represented by the Fourier series

1.8.1 Convergence of Coefficients and Parseval's Theorem

A different but related question concerns the convergence of the sequence of the Fourier coefficients themselves. First, we prove Bessel's inequality.

Theorem 1.8.2 (Bessel's Inequality). *Let f be a square-integrable function on $(-\pi, \pi)$ with Fourier coefficients:*

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx, \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx.$$

Then the following inequality holds:

$$\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \leq \frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 dx.$$

Proof. Consider the N -th partial sum of the Fourier series:

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^N (a_n \cos(nx) + b_n \sin(nx)).$$

Define the mean square error:

$$E_N = \int_{-\pi}^{\pi} [f(x) - S_N(x)]^2 dx.$$

Expanding the square:

$$E_N = \int_{-\pi}^{\pi} [f(x)]^2 dx - 2 \int_{-\pi}^{\pi} f(x) S_N(x) dx + \int_{-\pi}^{\pi} [S_N(x)]^2 dx.$$

We compute each term separately.

First, using the orthogonality of the trigonometric system:

$$\int_{-\pi}^{\pi} S_N(x) f(x) dx = \frac{a_0}{2} \int_{-\pi}^{\pi} f(x) dx + \sum_{n=1}^N \left[a_n \int_{-\pi}^{\pi} f(x) \cos(nx) dx + b_n \int_{-\pi}^{\pi} f(x) \sin(nx) dx \right].$$

Substituting the definitions of the Fourier coefficients:

$$\int_{-\pi}^{\pi} f(x) dx = \pi a_0, \quad \int_{-\pi}^{\pi} f(x) \cos(nx) dx = \pi a_n, \quad \int_{-\pi}^{\pi} f(x) \sin(nx) dx = \pi b_n,$$

we obtain:

$$\int_{-\pi}^{\pi} S_N(x) f(x) dx = \frac{a_0}{2} (\pi a_0) + \sum_{n=1}^N [a_n (\pi a_n) + b_n (\pi b_n)] = \pi \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right).$$

Next, compute the norm of $S_N(x)$:

$$\int_{-\pi}^{\pi} [S_N(x)]^2 dx = \int_{-\pi}^{\pi} \left[\frac{a_0}{2} + \sum_{n=1}^N (a_n \cos(nx) + b_n \sin(nx)) \right]^2 dx.$$

Again using orthogonality:

$$\int_{-\pi}^{\pi} \cos(nx) \cos(mx) dx = \begin{cases} \pi & \text{if } n = m \neq 0, \\ 2\pi & \text{if } n = m = 0, \\ 0 & \text{otherwise,} \end{cases} \quad \int_{-\pi}^{\pi} \sin(nx) \sin(mx) dx = \begin{cases} \pi & \text{if } n = m \neq 0, \\ 0 & \text{otherwise,} \end{cases}$$

and all cross terms vanish. Therefore:

$$\int_{-\pi}^{\pi} [S_N(x)]^2 dx = \pi \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right).$$

Substituting these into the expression for E_N :

$$E_N = \int_{-\pi}^{\pi} [f(x)]^2 dx - 2\pi \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right) + \pi \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right) = \int_{-\pi}^{\pi} [f(x)]^2 dx - \pi \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right).$$

Since $E_N \geq 0$, we conclude:

$$\pi \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right) \leq \int_{-\pi}^{\pi} [f(x)]^2 dx,$$

or equivalently:

$$\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \leq \frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 dx.$$

This holds for all N , so taking the limit as $N \rightarrow \infty$ yields Bessel's inequality:

$$\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \leq \frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 dx.$$

Theorem 1.8.3 (Convergence of Fourier Coefficients). *If a function $f(x)$ is bounded and integrable on the interval $(-\pi, \pi)$ with Fourier coefficients a_n and b_n , then the series formed by the sum of the squares of the coefficients converges. That is,*

$$\sum_{n=1}^{\infty} (a_n^2 + b_n^2) \quad \text{converges.}$$

Proof. By Bessel's inequality, for any positive integer N , the Fourier coefficients of f satisfy:

$$\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \leq \frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 dx.$$

Since f is bounded and integrable on $(-\pi, \pi)$, the function $[f(x)]^2$ is also integrable, so the right-hand side is finite. The partial sums

$$S_N = \sum_{n=1}^N (a_n^2 + b_n^2)$$

form a non-decreasing sequence bounded above by $\frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 dx$. Hence, the series $\sum_{n=1}^{\infty} (a_n^2 + b_n^2)$ converges. $\blacksquare$

Remark 1.8.4 (Riemann-Lebesgue Lemma). A necessary condition for the convergence of the series $\sum (a_n^2 + b_n^2)$ is that the terms must approach zero. This leads to the **Riemann-Lebesgue Lemma**, which states that if $f(x)$ is integrable on an interval, then its Fourier coefficients must tend to zero as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} a_n = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} b_n = 0$$

Theorem 1.8.5 (Riemann-Lebesgue Lemma). *Let f be an absolutely integrable function on the interval $[a, b]$, i.e.,*

$$\int_a^b |f(t)| dt < \infty.$$

Then:

$$\lim_{n \rightarrow \infty} \int_a^b f(t) \sin(nt) dt = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \int_a^b f(t) \cos(nt) dt = 0.$$

In particular, for Fourier coefficients of an integrable periodic function, we have:

$$\lim_{n \rightarrow \infty} a_n = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} b_n = 0.$$

Proof. We prove the result for the sine integral; the cosine case is analogous.

First consider a step function $\phi(t)$ on $[a, b]$, which is constant on subintervals:

$$\phi(t) = c_j \quad \text{for } t \in (t_{j-1}, t_j), \quad j = 1, \dots, m.$$

Then:

$$\int_a^b \phi(t) \sin(nt) dt = \sum_{j=1}^m c_j \int_{t_{j-1}}^{t_j} \sin(nt) dt = \sum_{j=1}^m \frac{c_j}{n} [-\cos(nt)]_{t_{j-1}}^{t_j}.$$

Since $|\cos(nt)| \leq 1$, we have:

$$\left| \int_a^b \phi(t) \sin(nt) dt \right| \leq \sum_{j=1}^m \frac{2|c_j|}{n} \leq \frac{2M}{n} \sum_{j=1}^m (t_j - t_{j-1}) = \frac{2M(b-a)}{n},$$

where $M = \max_j |c_j|$. Clearly, this tends to 0 as $n \rightarrow \infty$.

For any integrable function f and any $\epsilon > 0$, there exists a step function ϕ such that:

$$\int_a^b |f(t) - \phi(t)| dt < \frac{\epsilon}{2}.$$

Then:

$$\left| \int_a^b f(t) \sin(nt) dt \right| \leq \left| \int_a^b [f(t) - \phi(t)] \sin(nt) dt \right| + \left| \int_a^b \phi(t) \sin(nt) dt \right|.$$

The first term is bounded by:

$$\int_a^b |f(t) - \phi(t)| dt < \frac{\epsilon}{2}.$$

For the second term, since ϕ is a step function, there exists n_0 such that for all $n > n_0$:

$$\left| \int_a^b \phi(t) \sin(nt) dt \right| < \frac{\epsilon}{2}.$$

Hence, for $n > n_0$:

$$\left| \int_a^b f(t) \sin(nt) dt \right| < \epsilon,$$

which proves the result.

The application to Fourier coefficients follows by taking:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt) dt, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt,$$

and applying the lemma with $n = n$. ■

A stronger result, which turns Bessel's inequality into an equality, holds for functions satisfying Dirichlet's conditions.

Theorem 1.8.6 (Parseval's Theorem). *Let f be a periodic function with period $2L$ that satisfies Dirichlet's conditions, with Fourier series expansion:*

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right).$$

Then the following identity holds:

$$\frac{1}{L} \int_{-L}^L [f(x)]^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

Proof. Let $S_N(x)$ be the N -th partial sum of the Fourier series:

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^N \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right).$$

Consider the mean square error:

$$E_N = \int_{-L}^L [f(x) - S_N(x)]^2 dx.$$

Expanding the square:

$$E_N = \int_{-L}^L [f(x)]^2 dx - 2 \int_{-L}^L f(x) S_N(x) dx + \int_{-L}^L [S_N(x)]^2 dx.$$

We now compute each term separately.

First term:

$$I_1 = \int_{-L}^L [f(x)]^2 dx.$$

Second term: Using the definition of Fourier coefficients:

$$\int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx = La_n, \quad \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx = Lb_n,$$

and

$$\int_{-L}^L f(x) dx = La_0,$$

we have:

$$\int_{-L}^L f(x) S_N(x) dx = \frac{a_0}{2} (La_0) + \sum_{n=1}^N [a_n (La_n) + b_n (Lb_n)] = L \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right).$$

Third term: Using the orthogonality relations:

$$\int_{-L}^L \cos\left(\frac{n\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx = \begin{cases} L & \text{if } n = m \neq 0, \\ 2L & \text{if } n = m = 0, \\ 0 & \text{otherwise,} \end{cases}$$

$$\int_{-L}^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx = \begin{cases} L & \text{if } n = m \neq 0, \\ 0 & \text{otherwise,} \end{cases}$$

and all cross terms vanish. Therefore:

$$\int_{-L}^L [S_N(x)]^2 dx = L \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right).$$

Substituting into the expression for E_N :

$$E_N = I_1 - 2L \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right) + L \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right) = I_1 - L \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right).$$

Since $E_N \geq 0$, we obtain Bessel's inequality:

$$L \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right) \leq \int_{-L}^L [f(x)]^2 dx.$$

Under Dirichlet's conditions, the Fourier series converges in the mean, i.e., $\lim_{N \rightarrow \infty} E_N = 0$. Taking the limit as $N \rightarrow \infty$:

$$\lim_{N \rightarrow \infty} E_N = \int_{-L}^L [f(x)]^2 dx - L \left(\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right) = 0.$$

Thus,

$$\int_{-L}^L [f(x)]^2 dx = L \left(\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right),$$

or equivalently,

$$\frac{1}{L} \int_{-L}^L [f(x)]^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

■

Exercise 1.8.7. Solve the following problems related to Fourier series. Show all your work and justify each step.

1. Determine whether the following functions are even, odd, or neither:

(a) $f(x) = x^3 \cos(x)$

(b) $g(x) = e^{-|x|}$

(c) $h(x) = \sin(x^2)$

2. A square wave function is defined as:

$$f(x) = \begin{cases} 1 & \text{for } 0 < x < \pi \\ -1 & \text{for } -\pi < x < 0 \end{cases}$$

with period 2π . Find its Fourier series representation.

3. A sawtooth wave is given by:

$$f(x) = x \quad \text{for } -\pi < x < \pi$$

with period 2π . Determine its Fourier series expansion.

4. A triangular wave is defined as:

$$f(x) = \begin{cases} \pi + x & \text{for } -\pi \leq x \leq 0 \\ \pi - x & \text{for } 0 \leq x \leq \pi \end{cases}$$

with period 2π . Find its Fourier series.

5. For the function $f(x) = x^2$ on the interval $(-\pi, \pi)$:

(a) Verify Dirichlet's conditions

(b) Compute the Fourier coefficients a_n and b_n

(c) Write the complete Fourier series

(d) Using Parseval's identity, show that $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$

6. Consider the function $f(x) = e^x$ defined on $[0, \pi]$:

(a) Find its Fourier cosine series expansion

(b) Find its Fourier sine series expansion

(c) Compare the convergence rates of both series

7. For the function $f(x) = \cos(x)$ on $[0, \pi]$:

(a) Find the half-range sine series

(b) Find the half-range cosine series

(c) Explain why the results differ

8. Using the Riemann-Lebesgue lemma, show that for any absolutely integrable function $f(x)$ on $[-\pi, \pi]$, the Fourier coefficients satisfy:

$$\lim_{n \rightarrow \infty} a_n = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} b_n = 0$$

9. Consider the function:

$$f(x) = \begin{cases} x & \text{for } 0 \leq x \leq 1 \\ 2 - x & \text{for } 1 < x \leq 2 \end{cases}$$

with period 2. Find its Fourier series representation.

10. For the function $f(x) = \sin(\pi x)$ on $[-1, 1]$:

- (a) Find its Fourier series expansion
- (b) Verify that the series converges to the function at all points
- (c) Using appropriate x values, find the sums:

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{4n^2 - 1} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{4n^2 - 1}$$

Chapter 2

The Fourier Transform

The Fourier transform is a mathematical operation that transforms a function of time (or space) into a function of frequency. It decomposes a function into its constituent frequencies, similar to how a musical chord can be expressed as the frequencies of its constituent notes.

2.1 Definition and Basic Formulas

2.1.1 Fourier Transform

For a function $f(t)$ that is absolutely integrable on $(-\infty, \infty)$, the Fourier transform is defined as:

$$\mathcal{F}\{f(t)\} = F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt$$

where ω is the angular frequency.

Example 2.1.1 (Rectangular Pulse). Find the Fourier transform of the rectangular pulse:

$$f(t) = \begin{cases} 1 & \text{for } |t| < a \\ 0 & \text{for } |t| > a \end{cases}$$

Solution.

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt = \int_{-a}^a e^{-i\omega t} dt = \left[\frac{e^{-i\omega t}}{-i\omega} \right]_{-a}^a = \frac{2 \sin(a\omega)}{\omega}$$

This is the well-known unnormalized sinc function: $F(\omega) = 2a \cdot \text{sinc}(a\omega)$ ■

Example 2.1.2 (Gaussian Function). Find the Fourier transform of $f(t) = e^{-at^2}$.

Solution.

$$F(\omega) = \int_{-\infty}^{\infty} e^{-at^2} e^{-i\omega t} dt = \int_{-\infty}^{\infty} e^{-a(t^2 + \frac{i\omega}{a}t)} dt$$

Complete the square:

$$t^2 + \frac{i\omega}{a}t = \left(t + \frac{i\omega}{2a}\right)^2 - \frac{\omega^2}{4a^2}$$

Thus,

$$F(\omega) = e^{-\frac{\omega^2}{4a}} \int_{-\infty}^{\infty} e^{-a(t + \frac{i\omega}{2a})^2} dt = \sqrt{\frac{\pi}{a}} e^{-\frac{\omega^2}{4a}}$$
 ■

Example 2.1.3 (Exponential Decay). Find the Fourier transform of $f(t) = e^{-a|t|}$.

Solution.

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} e^{-a|t|} e^{-i\omega t} dt = \int_{-\infty}^0 e^{at} e^{-i\omega t} dt + \int_0^{\infty} e^{-at} e^{-i\omega t} dt \\ &= \int_{-\infty}^0 e^{(a-i\omega)t} dt + \int_0^{\infty} e^{-(a+i\omega)t} dt = \frac{1}{a-i\omega} + \frac{1}{a+i\omega} = \frac{2a}{a^2 + \omega^2} \end{aligned}$$
 ■

Example 2.1.4. Consider the rectangular pulse function defined as:

$$f(t) = \begin{cases} 1 & \text{for } -\frac{1}{2} \leq t \leq \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

The Fourier transform of $f(t)$ is given by:

$$F(\omega) = \mathcal{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt$$

Since $f(t)$ is zero outside the interval $[-\frac{1}{2}, \frac{1}{2}]$, we can write:

$$F(\omega) = \int_{-1/2}^{1/2} e^{-i\omega t} dt$$

Evaluating this integral:

$$F(\omega) = \left[\frac{e^{-i\omega t}}{-i\omega} \right]_{-1/2}^{1/2} = \frac{1}{-i\omega} (e^{-i\omega/2} - e^{i\omega/2})$$

Using Euler's formula, we can express this as:

$$F(\omega) = \frac{1}{-i\omega} (-2i \sin(\omega/2)) = \frac{2}{\omega} \sin\left(\frac{\omega}{2}\right)$$

This can be written in terms of the sinc function:

$$F(\omega) = \frac{\sin(\omega/2)}{\omega/2} = \text{sinc}\left(\frac{\omega}{2}\right)$$

where the sinc function is defined as:

$$\text{sinc}(x) = \frac{\sin x}{x}$$

Thus, the Fourier transform of the rectangular pulse is:

$$\boxed{\mathcal{F}\{\text{rect}(t)\} = \text{sinc}\left(\frac{\omega}{2}\right)}$$

Definition 2.1.5 (Inverse Fourier Transform). The inverse Fourier transform is given by:

$$\mathcal{F}^{-1}\{F(\omega)\} = f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{i\omega t} d\omega$$

2.1.2 Fourier Sine and Cosine Transforms

For real-valued functions, we often use sine and cosine transforms:

Fourier Cosine Transform:

$$\mathcal{F}_c\{f(t)\} = F_c(\omega) = \int_0^{\infty} f(t) \cos(\omega t) dt$$

Fourier Sine Transform:

$$\mathcal{F}_s\{f(t)\} = F_s(\omega) = \int_0^{\infty} f(t) \sin(\omega t) dt$$

2.2 Properties of Fourier Transform

1. **Linearity:** $\mathcal{F}\{af(t) + bg(t)\} = aF(\omega) + bG(\omega)$
2. **Time Scaling:** $\mathcal{F}\{f(at)\} = \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$
3. **Time Shifting:** $\mathcal{F}\{f(t - t_0)\} = e^{-i\omega t_0} F(\omega)$
4. **Frequency Shifting:** $\mathcal{F}\{e^{i\omega_0 t} f(t)\} = F(\omega - \omega_0)$
5. **Symmetry:** If $F(\omega)$ is the Fourier transform of $f(t)$, then $\mathcal{F}\{F(t)\} = 2\pi f(-\omega)$
6. **Derivative:** $\mathcal{F}\{f'(t)\} = i\omega F(\omega)$
7. **Convolution Theorem:** $\mathcal{F}\{f(t) * g(t)\} = F(\omega)G(\omega)$

Example 2.2.1. Let $f(t) = e^{-t^2}$ (Gaussian function). We know:

$$\mathcal{F}\{e^{-t^2}\} = \sqrt{\pi}e^{-\omega^2/4}$$

For $a = 2$, we have:

$$\mathcal{F}\{e^{-(2t)^2}\} = \mathcal{F}\{e^{-4t^2}\} = \frac{1}{2}\sqrt{\pi}e^{-(\omega/2)^2/4} = \frac{\sqrt{\pi}}{2}e^{-\omega^2/16}$$

Example 2.2.2. Let $f(t) = \text{rect}(t)$ (rectangular pulse from $-1/2$ to $1/2$). We know:

$$\mathcal{F}\{\text{rect}(t)\} = \text{sinc}(\omega/2\pi)$$

For $t_0 = 2$, we have:

$$\mathcal{F}\{\text{rect}(t - 2)\} = e^{-i2\omega}\text{sinc}(\omega/2\pi)$$

The magnitude remains unchanged while the phase is shifted by -2ω .

Example 2.2.3. Let $f(t) = \text{rect}(t)$ and $\omega_0 = 4\pi$. Then:

$$\mathcal{F}\{e^{i4\pi t}\text{rect}(t)\} = \text{sinc}((\omega - 4\pi)/2\pi)$$

This represents a frequency-shifted version of the sinc function.

Example 2.2.4. Let $f(t) = e^{-|t|}$. We know:

$$\mathcal{F}\{e^{-|t|}\} = \frac{2}{1 + \omega^2}$$

Then:

$$\mathcal{F}\left\{\frac{2}{1 + t^2}\right\} = 2\pi e^{-|\omega|} = 2\pi e^{-|\omega|}$$

Example 2.2.5. Let $f(t) = e^{-t}u(t)$. Then $f'(t) = -e^{-t}u(t) + \delta(t)$. We know:

$$\mathcal{F}\{e^{-t}u(t)\} = \frac{1}{1 + i\omega}$$

Using the derivative property:

$$\mathcal{F}\{f'(t)\} = i\omega \cdot \frac{1}{1 + i\omega} = \frac{i\omega}{1 + i\omega}$$

Example 2.2.6. $\mathcal{F}\{f(t) * g(t)\} = F(\omega)G(\omega)$

Example: Let $f(t) = g(t) = \text{rect}(t)$. The convolution of two rectangular pulses is a triangular pulse:

$$(f * g)(t) = \begin{cases} 1 - |t| & |t| < 1 \\ 0 & |t| \geq 1 \end{cases}$$

We know:

$$\mathcal{F}\{\text{rect}(t)\} = \text{sinc}(\omega/2\pi)$$

By the convolution theorem:

$$\mathcal{F}\{f(t) * g(t)\} = [\text{sinc}(\omega/2\pi)]^2$$

Now, we prove these properties of Fourier transform one by one.

Theorem 2.2.7 (Linearity). For any constants a and b , and functions $f(t)$ and $g(t)$ with Fourier transforms $F(\omega)$ and $G(\omega)$ respectively:

$$\mathcal{F}\{af(t) + bg(t)\} = aF(\omega) + bG(\omega)$$

Proof. By the definition of the Fourier transform:

$$\mathcal{F}\{af(t) + bg(t)\} = \int_{-\infty}^{\infty} [af(t) + bg(t)]e^{-i\omega t} dt$$

Using the linearity of integration:

$$= a \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt + b \int_{-\infty}^{\infty} g(t)e^{-i\omega t} dt = aF(\omega) + bG(\omega)$$

■

Theorem 2.2.8 (Time Scaling). For a constant $a \neq 0$:

$$\mathcal{F}\{f(at)\} = \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$$

Proof. Consider first the case $a > 0$. Let $u = at$, then $du = a dt$, $dt = du/a$:

$$\mathcal{F}\{f(at)\} = \int_{-\infty}^{\infty} f(at)e^{-i\omega t} dt = \int_{-\infty}^{\infty} f(u)e^{-i\omega(u/a)} \frac{du}{a} = \frac{1}{a} \int_{-\infty}^{\infty} f(u)e^{-i(\omega/a)u} du = \frac{1}{a} F\left(\frac{\omega}{a}\right)$$

For $a < 0$, let $a = -|a|$. Then $u = at = -|a|t$, $du = -|a|dt$, $dt = -du/|a|$:

$$\mathcal{F}\{f(at)\} = \int_{-\infty}^{\infty} f(at)e^{-i\omega t} dt = \int_{\infty}^{-\infty} f(u)e^{-i\omega(u/a)} \left(-\frac{du}{|a|}\right) = \frac{1}{|a|} \int_{-\infty}^{\infty} f(u)e^{-i(\omega/a)u} du = \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$$

Combining both cases gives the result:

$$\mathcal{F}\{f(at)\} = \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$$

■

Theorem 2.2.9 (Time Shifting). *For any real constant t_0 :*

$$\mathcal{F}\{f(t - t_0)\} = e^{-i\omega t_0} F(\omega)$$

Proof. Let $u = t - t_0$, then $t = u + t_0$, $dt = du$:

$$\begin{aligned} \mathcal{F}\{f(t - t_0)\} &= \int_{-\infty}^{\infty} f(t - t_0)e^{-i\omega t} dt = \int_{-\infty}^{\infty} f(u)e^{-i\omega(u+t_0)} du \\ &= e^{-i\omega t_0} \int_{-\infty}^{\infty} f(u)e^{-i\omega u} du = e^{-i\omega t_0} F(\omega) \end{aligned}$$

■

Theorem 2.2.10 (Frequency Shifting). *For any real constant ω_0 :*

$$\mathcal{F}\{e^{i\omega_0 t} f(t)\} = F(\omega - \omega_0)$$

Proof.

$$\mathcal{F}\{e^{i\omega_0 t} f(t)\} = \int_{-\infty}^{\infty} e^{i\omega_0 t} f(t)e^{-i\omega t} dt = \int_{-\infty}^{\infty} f(t)e^{-i(\omega - \omega_0)t} dt = F(\omega - \omega_0)$$

■

Theorem 2.2.11 (Symmetry). *If $F(\omega)$ is the Fourier transform of $f(t)$, then:*

$$\mathcal{F}\{F(t)\} = 2\pi f(-\omega)$$

Proof. By definition of the Fourier transform and its inverse:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{i\omega t} d\omega$$

Replace t with $-t$:

$$f(-t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{-i\omega t} d\omega$$

Now, by definition of the Fourier transform:

$$\mathcal{F}\{F(t)\} = \int_{-\infty}^{\infty} F(t)e^{-i\omega t} dt$$

Comparing these two expressions, we see that:

$$\mathcal{F}\{F(t)\} = 2\pi f(-\omega)$$

This completes the proof.

■

Theorem 2.2.12 (Derivative). *If $f(t)$ has Fourier transform $F(\omega)$, then:*

$$\mathcal{F}\{f'(t)\} = i\omega F(\omega)$$

Proof. By definition:

$$\mathcal{F}\{f'(t)\} = \int_{-\infty}^{\infty} f'(t)e^{-i\omega t} dt$$

Using integration by parts with:

$$\begin{aligned} u &= e^{-i\omega t} & dv &= f'(t)dt \\ du &= -i\omega e^{-i\omega t} dt & v &= f(t) \end{aligned}$$

we get:

$$\mathcal{F}\{f'(t)\} = [f(t)e^{-i\omega t}]_{-\infty}^{\infty} + i\omega \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt$$

Since $f(t)$ must vanish at $\pm\infty$ for the Fourier transform to exist, the boundary term is zero. Thus:

$$\mathcal{F}\{f'(t)\} = i\omega \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt = i\omega F(\omega)$$

This completes the proof. ■

Remark 2.2.13 (The general Case). If $\mathcal{F}[f(t)] = F(\omega)$, then $\mathcal{F}^n[f(t)] = (i\omega)^n F(\omega)$, where $f(x)$ and all its derivatives vanish at ∞ .

Theorem 2.2.14 (Convolution Theorem). If $f(t)$ and $g(t)$ have Fourier transforms $F(\omega)$ and $G(\omega)$ respectively, then:

$$\mathcal{F}\{f(t) * g(t)\} = F(\omega)G(\omega)$$

where the convolution is defined as:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t - \tau)d\tau$$

Proof. By definition:

$$\mathcal{F}\{f(t) * g(t)\} = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(\tau)g(t - \tau)d\tau \right] e^{-i\omega t} dt$$

Interchanging the order of integration (justified by Fubini's theorem):

$$= \int_{-\infty}^{\infty} f(\tau) \left[\int_{-\infty}^{\infty} g(t - \tau)e^{-i\omega t} dt \right] d\tau$$

Make the substitution $u = t - \tau$ in the inner integral:

$$\begin{aligned} &= \int_{-\infty}^{\infty} f(\tau) \left[\int_{-\infty}^{\infty} g(u)e^{-i\omega(u+\tau)} du \right] d\tau \\ &= \int_{-\infty}^{\infty} f(\tau)e^{-i\omega\tau} d\tau \cdot \int_{-\infty}^{\infty} g(u)e^{-i\omega u} du = F(\omega)G(\omega) \end{aligned}$$

This completes the proof. ■

Theorem 2.2.15 (Modulation Property). If $F(\omega)$ is the Fourier transform of $f(t)$, then for any real constant ω_0 :

$$(i) \quad \mathcal{F}\{f(t) \cos(\omega_0 t)\} = \frac{1}{2}[F(\omega + \omega_0) + F(\omega - \omega_0)]$$

$$(ii) \quad \mathcal{F}\{f(t) \sin(\omega_0 t)\} = \frac{i}{2}[F(\omega + \omega_0) - F(\omega - \omega_0)]$$

Proof. (i) We begin by expressing the cosine function in terms of complex exponentials using Euler's formula:

$$\cos(\omega_0 t) = \frac{e^{i\omega_0 t} + e^{-i\omega_0 t}}{2}$$

Now, applying the Fourier transform to the modulated function:

$$\begin{aligned} \mathcal{F}\{f(t) \cos(\omega_0 t)\} &= \int_{-\infty}^{\infty} f(t) \cos(\omega_0 t) e^{-i\omega t} dt \\ &= \int_{-\infty}^{\infty} f(t) \left(\frac{e^{i\omega_0 t} + e^{-i\omega_0 t}}{2} \right) e^{-i\omega t} dt \\ &= \frac{1}{2} \int_{-\infty}^{\infty} f(t) e^{i\omega_0 t} e^{-i\omega t} dt + \frac{1}{2} \int_{-\infty}^{\infty} f(t) e^{-i\omega_0 t} e^{-i\omega t} dt \\ &= \frac{1}{2} \int_{-\infty}^{\infty} f(t) e^{-i(\omega - \omega_0)t} dt + \frac{1}{2} \int_{-\infty}^{\infty} f(t) e^{-i(\omega + \omega_0)t} dt \\ &= \frac{1}{2} F(\omega - \omega_0) + \frac{1}{2} F(\omega + \omega_0) \end{aligned}$$
■

Definition 2.2.16. The unit step function $u_0(t)$ is defined as

$$u_0(t) = \begin{cases} 0; & t < 0 \\ 1; & t > 0 \end{cases}$$

Example 2.2.17. Find the solution of the differential equation

$$y' - 2y = u_0(t)e^{-2t}, -\infty < t < \infty.$$

using the Fourier transform.

Solution. Let $\hat{y}(\omega) = \mathcal{F}\{y(t)\}(\omega)$. Using the property $\mathcal{F}\{y'(t)\}(\omega) = i\omega\hat{y}(\omega)$, we obtain:

$$(i\omega - 2)\hat{y}(\omega) = \mathcal{F}\{u_0(t)e^{-2t}\}(\omega). \quad (1)$$

Now,

$$\mathcal{F}\{u_0(t)e^{-2t}\}(\omega) = \int_0^{\infty} e^{-(2+i\omega)t} dt = \frac{1}{2+i\omega}.$$

Therefore,

$$(i\omega - 2)\hat{y}(\omega) = \frac{1}{2+i\omega} \Rightarrow \hat{y}(\omega) = \frac{1}{(i\omega - 2)(2+i\omega)}.$$

Note that $2+i\omega = i\omega+2$, so:

$$\hat{y}(\omega) = \frac{1}{(i\omega - 2)(i\omega + 2)} = \frac{1}{(i\omega)^2 - 4} = -\frac{1}{\omega^2 + 4}.$$

We know that,

$$\mathcal{F}\left\{e^{-2|t|}\right\}(\omega) = \frac{4}{\omega^2 + 4} \Rightarrow \mathcal{F}\left\{\frac{1}{4}e^{-2|t|}\right\}(\omega) = \frac{1}{\omega^2 + 4}.$$

Thus,

$$\hat{y}(\omega) = -\frac{1}{\omega^2 + 4} = -\mathcal{F}\left\{\frac{1}{4}e^{-2|t|}\right\}(\omega) = \mathcal{F}\left\{-\frac{1}{4}e^{-2|t|}\right\}(\omega),$$

so the inverse transform gives:

$$y(t) = -\frac{1}{4}e^{-2|t|}.$$

Verification

- For $t > 0$: $y(t) = -\frac{1}{4}e^{-2t}$, $y'(t) = \frac{1}{2}e^{-2t}$, and

$$y' - 2y = \frac{1}{2}e^{-2t} + \frac{1}{2}e^{-2t} = e^{-2t}.$$

- For $t < 0$: $y(t) = -\frac{1}{4}e^{2t}$, $y'(t) = -\frac{1}{2}e^{2t}$, and

$$y' - 2y = -\frac{1}{2}e^{2t} + \frac{1}{2}e^{2t} = 0.$$

The solution satisfies the equation for all t .

Theorem 2.2.18. Let $f(t)$ be piecewise continuous on every interval $[-l, l]$. Suppose that

$$\int_{-\infty}^{\infty} |t^n f(t)| dt$$

converges. Then

$$\mathcal{F}[t^n f(t)](\omega) = i^n \frac{d^n}{d\omega^n} F(\omega),$$

where $F(\omega) = \mathcal{F}[f(t)](\omega)$ is the Fourier transform of $f(t)$.

Proof. We prove the result by induction on n .

For $n = 0$, the statement is trivial since $\mathcal{F}[f(t)](\omega) = F(\omega) = i^0 F^{(0)}(\omega)$.

Now, for $n = 1$, we wish to show that

$$\mathcal{F}[tf(t)](\omega) = i \frac{d}{d\omega} F(\omega).$$

Recall that

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt.$$

Under the assumption that $\int_{-\infty}^{\infty} |tf(t)| dt$ converges, we may differentiate under the integral sign. Specifically, consider the derivative with respect to ω :

$$\frac{d}{d\omega} F(\omega) = \frac{d}{d\omega} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt.$$

Note that

$$\frac{\partial}{\partial \omega} (f(t) e^{-i\omega t}) = -it f(t) e^{-i\omega t},$$

and

$$|-it f(t) e^{-i\omega t}| = |t f(t)|.$$

Since $\int_{-\infty}^{\infty} |t f(t)| dt$ converges, the dominated convergence theorem (or Leibniz's rule for differentiation under the integral sign) applies, and we have

$$\frac{d}{d\omega} F(\omega) = \int_{-\infty}^{\infty} \frac{\partial}{\partial \omega} (f(t) e^{-i\omega t}) dt = \int_{-\infty}^{\infty} -it f(t) e^{-i\omega t} dt = -i \int_{-\infty}^{\infty} t f(t) e^{-i\omega t} dt.$$

Therefore,

$$\int_{-\infty}^{\infty} t f(t) e^{-i\omega t} dt = i \frac{d}{d\omega} F(\omega),$$

which is exactly

$$\mathcal{F}[tf(t)](\omega) = i F'(\omega).$$

Now assume that the result holds for $n - 1$, i.e.,

$$\mathcal{F}[t^{n-1} f(t)](\omega) = i^{n-1} F^{(n-1)}(\omega).$$

Define $g(t) = t^{n-1} f(t)$. Then $t^n f(t) = t g(t)$. Note that $g(t)$ is piecewise continuous on every finite interval, and by hypothesis,

$$\int_{-\infty}^{\infty} |t g(t)| dt = \int_{-\infty}^{\infty} |t^n f(t)| dt$$

converges. Therefore, we may apply the $n = 1$ case to $g(t)$ to obtain

$$\mathcal{F}[t g(t)](\omega) = i \frac{d}{d\omega} \mathcal{F}[g(t)](\omega).$$

But by the induction hypothesis,

$$\mathcal{F}[g(t)](\omega) = i^{n-1} F^{(n-1)}(\omega),$$

so

$$\mathcal{F}[t^n f(t)](\omega) = i \frac{d}{d\omega} (i^{n-1} F^{(n-1)}(\omega)) = i^n F^{(n)}(\omega).$$

This completes the induction step, and the theorem is proved. ■

Theorem 2.2.19. Let $f(t)$ be piecewise continuous on every interval $[-l, l]$. Suppose that

$$\int_{-\infty}^{\infty} |f(t)| dt$$

converges. Let $\mathcal{F}[f(t)] = F(\omega)$ and suppose that $F(0) = 0$. Then

$$\mathcal{F}\left[\int_{-\infty}^t f(\tau) d\tau\right] = \frac{1}{i\omega} F(\omega).$$

Proof. Let the function $g(t)$ be defined as the integral of $f(t)$:

$$g(t) = \int_{-\infty}^t f(\tau) d\tau$$

Our goal is to compute the Fourier transform of $g(t)$, which we denote by $G(\omega)$. By the definition of the Fourier transform, we have:

$$G(\omega) = \mathcal{F}[g(t)] = \int_{-\infty}^{\infty} g(t) e^{-i\omega t} dt = \int_{-\infty}^{\infty} \left(\int_{-\infty}^t f(\tau) d\tau \right) e^{-i\omega t} dt$$

We can solve this integral using integration by parts, where $\int u dv = uv - \int v du$. Let's choose our parts as follows:

$$\begin{aligned} u &= \int_{-\infty}^t f(\tau) d\tau & dv &= e^{-i\omega t} dt \\ du &= f(t) dt & v &= \frac{e^{-i\omega t}}{-i\omega} \end{aligned}$$

Applying the integration by parts formula gives:

$$G(\omega) = \left[\left(\int_{-\infty}^t f(\tau) d\tau \right) \frac{e^{-i\omega t}}{-i\omega} \right]_{t=-\infty}^{t=\infty} - \int_{-\infty}^{\infty} \left(\frac{e^{-i\omega t}}{-i\omega} \right) f(t) dt$$

Now, we must evaluate the boundary term. Let's analyze it at the limits $t \rightarrow \infty$ and $t \rightarrow -\infty$.

1. **At the upper limit** ($t \rightarrow \infty$): The term is $\lim_{t \rightarrow \infty} \left(\int_{-\infty}^t f(\tau) d\tau \right) \frac{e^{-i\omega t}}{-i\omega}$. The integral part becomes:

$$\lim_{t \rightarrow \infty} \int_{-\infty}^t f(\tau) d\tau = \int_{-\infty}^{\infty} f(\tau) d\tau$$

We can relate this integral to the Fourier transform $F(\omega)$ evaluated at $\omega = 0$:

$$F(0) = \int_{-\infty}^{\infty} f(t) e^{-i(0)t} dt = \int_{-\infty}^{\infty} f(t) dt$$

The theorem gives the crucial condition that $F(0) = 0$. Therefore, $\int_{-\infty}^{\infty} f(\tau) d\tau = 0$. Since the integral part of the boundary term goes to 0 and the term $\frac{e^{-i\omega t}}{-i\omega}$ is bounded, the entire expression evaluates to 0 at the upper limit.

2. **At the lower limit** ($t \rightarrow -\infty$): The term is $\lim_{t \rightarrow -\infty} \left(\int_{-\infty}^t f(\tau) d\tau \right) \frac{e^{-i\omega t}}{-i\omega}$. The integral part becomes:

$$\lim_{t \rightarrow -\infty} \int_{-\infty}^t f(\tau) d\tau = 0$$

by the definition of an improper integral converging. Thus, the boundary term is also 0 at the lower limit.

Since the boundary term vanishes completely, we are left with only the integral part from our integration by parts formula:

$$\begin{aligned} G(\omega) &= - \int_{-\infty}^{\infty} \left(\frac{e^{-i\omega t}}{-i\omega} \right) f(t) dt \\ &= \frac{1}{i\omega} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \end{aligned}$$

The remaining integral is, by definition, the Fourier transform of $f(t)$, which is $F(\omega)$. Therefore, we have shown that:

$$G(\omega) = \mathcal{F} \left[\int_{-\infty}^t f(\tau) d\tau \right] = \frac{1}{i\omega} F(\omega)$$

This completes the proof. ■

Example 2.2.20. Using the convolution theorem, find the inverse Fourier transform of the function

$$Y(\omega) = \frac{1}{12 + 7i\omega - \omega^2}$$

That is, find $y(t) = \mathcal{F}^{-1}[Y(\omega)]$.

Solution. The strategy is to use the convolution theorem in its "inverse" form:

$$\mathcal{F}^{-1}[F(\omega)G(\omega)] = (f * g)(t)$$

where $f(t) = \mathcal{F}^{-1}[F(\omega)]$ and $g(t) = \mathcal{F}^{-1}[G(\omega)]$. First, we must decompose the given function $Y(\omega)$ into a product of two simpler functions, $F(\omega)$ and $G(\omega)$. We do this by factoring the denominator.

Step 1: Factor the Denominator Let's rearrange the denominator and factor it. Notice that $-\omega^2 = (i\omega)^2$.

$$\begin{aligned} 12 + 7i\omega - \omega^2 &= 12 + 7(i\omega) + (i\omega)^2 \\ &= ((i\omega) + 3)((i\omega) + 4) \\ &= (3 + i\omega)(4 + i\omega) \end{aligned}$$

So, we can write $Y(\omega)$ as a product:

$$Y(\omega) = \frac{1}{(3 + i\omega)} \cdot \frac{1}{(4 + i\omega)}$$

Step 2: Identify the Individual Transforms and their Inverses We can now identify our two functions in the frequency domain:

$$F(\omega) = \frac{1}{3 + i\omega} \quad \text{and} \quad G(\omega) = \frac{1}{4 + i\omega}$$

We use the standard Fourier transform pair for a one-sided decaying exponential:

$$\mathcal{F}^{-1} \left[\frac{1}{a + i\omega} \right] = e^{-at} u(t)$$

where $u(t)$ is the Heaviside unit step function. Applying this rule to our functions, we find their corresponding time-domain functions:

$$\begin{aligned} f(t) &= \mathcal{F}^{-1}[F(\omega)] = e^{-3t} u(t) \\ g(t) &= \mathcal{F}^{-1}[G(\omega)] = e^{-4t} u(t) \end{aligned}$$

Step 3: Compute the Convolution According to the convolution theorem, the desired function $y(t)$ is the convolution of $f(t)$ and $g(t)$.

$$y(t) = (f * g)(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau$$

Substituting our functions:

$$y(t) = \int_{-\infty}^{\infty} [e^{-3\tau} u(\tau)] \cdot [e^{-4(t-\tau)} u(t - \tau)] d\tau$$

The Heaviside functions $u(\tau)$ and $u(t - \tau)$ constrain the limits of integration. The integrand is non-zero only when both $\tau \geq 0$ and $t - \tau \geq 0$ (which implies $\tau \leq t$). Therefore, the integration limits become 0 to t . This also implies that the result is zero for $t < 0$. For $t \geq 0$:

$$\begin{aligned} y(t) &= \int_0^t e^{-3\tau} e^{-4(t-\tau)} d\tau \\ &= \int_0^t e^{-3\tau} e^{-4t} e^{4\tau} d\tau \\ &= e^{-4t} \int_0^t e^{\tau} d\tau \\ &= e^{-4t} [e^{\tau}]_0^t \\ &= e^{-4t} (e^t - e^0) \\ &= e^{-4t} (e^t - 1) \\ &= e^{-3t} - e^{-4t} \end{aligned}$$

Since the result is valid only for $t \geq 0$, we multiply by the Heaviside function to get the final answer.

$$y(t) = (e^{-3t} - e^{-4t}) u(t)$$

■

Exercise 2.2.21. 1. Find the Fourier transform of $f(t) = \delta(t - t_0)$, where δ is the Dirac delta function.

2. Find the Fourier transform of $f(t) = \cos(\omega_0 t)$.

3. Using the convolution theorem, find the Fourier transform of the triangular function:

$$f(t) = \begin{cases} 1 - |t|/a & \text{for } |t| < a \\ 0 & \text{for } |t| > a \end{cases}$$

2.3 Discrete Fourier Transform (DFT)

2.3.1 Definition

The Discrete Fourier Transform (DFT) converts a finite sequence of equally-spaced samples of a function into a sequence of coefficients of a finite combination of complex sinusoids.

For a sequence of N complex numbers $x_0, x_1, \dots, x_{N-1}$, the DFT is defined as:

$$X_k = \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi kn/N} \quad \text{for } k = 0, 1, \dots, N-1$$

where:

- X_k is the k -th frequency component
- x_n is the n -th input sample
- N is the total number of samples
- i is the imaginary unit ($i^2 = -1$)

2.3.2 Inverse DFT

The inverse DFT reconstructs the original sequence from the frequency components:

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k \cdot e^{i2\pi kn/N} \quad \text{for } n = 0, 1, \dots, N-1$$

2.3.3 Properties of DFT

1. **Linearity:** For any sequences x_n and y_n and constants a and b :

$$\mathcal{F}\{ax_n + by_n\} = a\mathcal{F}\{x_n\} + b\mathcal{F}\{y_n\}$$

2. **Time Shifting:** Shifting the input sequence by m samples multiplies the DFT by a phase factor:

$$\mathcal{F}\{x_{n-m}\} = X_k \cdot e^{-i2\pi km/N}$$

3. **Frequency Shifting:** Multiplying the input by a complex exponential shifts the frequency components:

$$\mathcal{F}\{x_n \cdot e^{i2\pi mn/N}\} = X_{k-m}$$

4. **Convolution Theorem:** Circular convolution in time domain equals pointwise multiplication in frequency domain:

$$\mathcal{F}\{x_n * y_n\} = X_k \cdot Y_k$$

5. **Parseval's Theorem:** The total energy is conserved between time and frequency domains:

$$\sum_{n=0}^{N-1} |x_n|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X_k|^2$$

6. **Symmetry:** For real-valued input signals:

$$X_k = X_{N-k}^*$$

where $*$ denotes complex conjugation.

2.4 Fast Fourier Transform (FFT)

2.4.1 Introduction

The Fast Fourier Transform (FFT) is an algorithm that efficiently computes the DFT and its inverse. While a direct implementation of the DFT requires $O(N^2)$ operations, the FFT reduces this to $O(N \log N)$ operations.

2.4.2 Cooley-Tukey Algorithm (Radix-2 FFT)

The most common FFT algorithm is the Cooley-Tukey algorithm, which uses a divide-and-conquer approach. It works by recursively breaking down a DFT of size N into smaller DFTs.

$$\begin{aligned}
 X_k &= \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi kn/N} \\
 &= \sum_{m=0}^{N/2-1} x_{2m} \cdot e^{-i2\pi k(2m)/N} + \sum_{m=0}^{N/2-1} x_{2m+1} \cdot e^{-i2\pi k(2m+1)/N} \\
 &= \sum_{m=0}^{N/2-1} x_{2m} \cdot e^{-i2\pi km/(N/2)} + e^{-i2\pi k/N} \sum_{m=0}^{N/2-1} x_{2m+1} \cdot e^{-i2\pi km/(N/2)}
 \end{aligned}$$

This shows that the DFT can be computed as the sum of two smaller DFTs:

- One for the even-indexed elements
- One for the odd-indexed elements (with a twiddle factor)

2.4.3 FFT Algorithm Steps

1. If $N = 1$, return x_0 (base case)
2. Separate even and odd indices:

$$x_{\text{even}} = [x_0, x_2, \dots, x_{N-2}], \quad x_{\text{odd}} = [x_1, x_3, \dots, x_{N-1}]$$

3. Recursively compute DFTs of both halves:

$$X_{\text{even}} = \text{FFT}(x_{\text{even}}), \quad X_{\text{odd}} = \text{FFT}(x_{\text{odd}})$$

4. Combine the results using the twiddle factors $\omega_N^k = e^{-i2\pi k/N}$:

$$\begin{aligned}
 X_k &= X_{\text{even}}[k] + \omega_N^k \cdot X_{\text{odd}}[k] \\
 X_{k+N/2} &= X_{\text{even}}[k] - \omega_N^k \cdot X_{\text{odd}}[k] \quad \text{for } k = 0, 1, \dots, N/2 - 1
 \end{aligned}$$

2.4.4 Properties and Advantages of FFT

- **Speed:** Reduces computation time from $O(N^2)$ to $O(N \log N)$
- **Memory Efficiency:** Many implementations use in-place computation
- **Numerical Stability:** Generally well-conditioned algorithm
- **Flexibility:** Works for any N , though most efficient when N is a power of 2